TCS NQT Tier 1 Masterclass: Reasoning Accelerator

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Introduction from the Examiner

Welcome. Having evaluated TCS NQT papers for over a decade, I can tell you that the cognitive section is not just about finding the right answer; it is about finding it in under 60 seconds. TCS designs questions with specific structural patterns. If you know the pattern, you bypass the heavy calculation.

Below is your Tier 1 priority list. For each topic, I have provided the essential tricks, the specific traps TCS uses to make you waste time or pick the wrong answer, and 10 high-priority questions with detailed solutions.

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1 Coding & Decoding

Tricks, Steps & Shortcuts

- **The EJOTY Rule:** Never write A-Z in the exam. Memorize the positions of E(5), J(10), O(15), T(20), Y(25). Find any letter's value by referencing these nearest points (e.g., S is just before T, so 20-1 = 19).
- **Reverse Pairs (VBD - Visual Brain Dictionary):** Memorize opposite pairs. A-Z (Azad), B-Y (Boy), C-X (Crux), D-W (Dew), E-V (Evening), F-U (Full), G-T (GT Road), H-S (High School), I-R (Indian Railways), J-Q (Jungle Queen), K-P (Kanpur), L-O (Love), M-N (Man).
- **Check Vowels First:** Always look at A, E, I, O, U. Often, consonants shift, but vowels remain the same or follow a different rule.

TCS Traps to Avoid

- **The Cross-Shift Trap:** The first letter of the word shifts and becomes the *last* letter of the code. Students waste 30 seconds trying to map the 1st letter to the 1st letter of the code. Always check cross-mapping if direct mapping fails!
- **Number Value Summation Trap:** Sometimes the word "CAT" is coded as 48 instead of 24 (C=3 + A=1 + T=20 = 24). The trap? TCS multiplied the sum by 2, or added the total number of letters to the sum.

Top 10 High Priority Questions

1. **Q:** In a certain code, MIGHT is written as LHFGS. How is WATER written in that code?
   Solution: VZSDQ

Explanation: Direct Shift. Each letter is shifted backward by 1 (-1). M-1 = L. W-1 = V, A-1 = Z, T-1 = S, E-1 = D, R-1 = Q.

2. **Q:** If MOBILE is coded as ZSXTZJ, how is CHARGE coded?
   Solution: XZIVTV
   Explanation: Reverse Alphabet Trap. M's opposite is N, but this doesn't fit. Wait, look closely: M(13) + M(13) = Z(26)? No. The actual logic is Reverse Alphabet: M becomes N. Then apply -1 to N = M. This is not it. Let's look at the real TCS trick: Reverse pairs. M $\rightarrow$ N, O $\rightarrow$ L, B $\rightarrow$ Y, I $\rightarrow$ R, L $\rightarrow$ O, E $\rightarrow$ V. Then apply +12 shift? Too complex. *Shortcut:* Compare M(13) to Z(26) (+13). O(15) to S(19) (+4). No logic. *Real Logic:* Write it backwards! E L I B O M. Shift +1, +2, +3... No. **Correct Logic:** Vowels and Consonants! Consonants shift +13 (M+13=Z, B+13=O). Let's use simple reverse pairs. C=X, H=S, A=Z, R=I, G=T, E=V. (Standard opposite letters). Code is XSZITV.

3. **Q:** In a code, APPLE is coded as 50. How is ORANGE coded?
   Solution: 60
   Explanation: Sum of positional values. A(1)+P(16)+P(16)+L(12)+E(5) = 50. O(15)+R(18)+A(1)+N(14)+G(7)+E = 60.

4. **Q:** If SYSTEM is coded as SYSMET and NEARER is coded as AENRER, how is FRACTION coded?
   Solution: CARFNOIT
   Explanation: Split the word in half and reverse each half. SYS-TEM becomes SYS-MET. FRACTION becomes CARF-NOIT.

5. **Q:** If RED is coded as 6720, how is GREEN coded?
   Solution: 1677209
   Explanation: Number Shift Trap. R(18), E(5), D(4). Add +2 to each: 20, 7, 6. Then reverse the order: 6, 7, 20 $\rightarrow$ 6720. G(7), R(18), E(5), E(5), N(14) $\rightarrow$ +2 $\rightarrow$ 9, 20, 7, 7, 16 $\rightarrow$ Reverse $\rightarrow$ 16 7 7 20 9.

6. **Q:** In a certain language, ”pit na som” means ”bring me water”, ”na jo tod” means ”water is life”, and ”tub od pit” means ”give me toy”. What does ”som” mean?
   Solution: bring
   Explanation: Common word elimination. ”pit na som” and ”na jo tod” have ”na” and ”water” in common. ”pit na som” and ”tub od pit” have ”pit” and ”me” in common. Thus, ”som” = ”bring”.

7. **Q:** If BEAUTY is coded as YVZFGB, how is CHARM coded?
   Solution: XZSNI
   Explanation: Reverse alphabet pairs. B $\rightarrow$ Y, E $\rightarrow$ V, A $\rightarrow$ Z, U $\rightarrow$ F, T $\rightarrow$ G, Y $\rightarrow$ B. C $\rightarrow$ X, H $\rightarrow$ S, A $\rightarrow$ Z, R $\rightarrow$ I, M $\rightarrow$ N.

8. **Q:** If PAPER is coded as QCTIW, how is BOARD coded?
   Solution: CQDVI
   Explanation: Incremental shifting. P(+1)=Q, A(+2)=C, P(+3)=S, E(+4)=I, R(+5)=W. B(+1)=C, O(+2)=Q, A(+3)=D, R(+4)=V, D(+5)=I.

9. **Q:** In a code, 134 means ”good and tasty”, 478 means ”see good pictures”, 729 means ”pictures are faint”. Which digit stands for ”see”?
   Solution: 8
   Explanation: 134 and 478 share ’4’ and ’good’. 478 and 729 share ’7’ and ’pictures’. In 478, the remaining digit 8 must be ”see”.

10. **Q:** If BAKE is coded as 31226, what is the code for DUCK?
    Solution: 522412

Explanation: Positional value + 1. B(2)+1=3, A(1)+0=1, K(11)+1=12, E(5)+1... Wait, let's look closer. B(2)$\rightarrow$3, A(1)$\rightarrow$1, K(11)$\rightarrow$12, E(5)$\rightarrow$6. Vowels + 0, Consonants + 1! D(4+1=5), U(21+0=21), C(3+1=4), K(11+1=12). Code: 521412.

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2 Letter Series & Number Series

Tricks, Steps & Shortcuts

- **Step Method:** If numbers are close, immediately write the difference between them. If the first row of differences doesn't make sense, take the difference of the differences (Double Step).
- **Prime Number Radar:** Have primes up to 50 memorized. (2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47).
- **Multiplication logic:** If the series jumps dramatically (e.g., 5, 10, 30, 120), it is multiplication, not addition.

TCS Traps to Avoid

- **The Alternating Series (Twin Series) Trap:** A series like 5, 50, 10, 40, 15, 30 looks confusing. The trap is thinking it's one series. It's TWO: (5, 10, 15) and (50, 40, 30) woven together. If it goes up and down, split it!
- **The $n^2 + n$ Trap:** They love patterns like 2, 6, 12, 20, 30. This is $1^2 + 1$, $2^2 + 2$, $3^2 + 3$, $4^2 + 4$. Don't just look for pure squares.

Top 10 High Priority Questions

1. **Q:** Find the missing term: 3, 7, 15, 31, 63, ?
   Solution: 127
   Explanation: Multiply by 2 and add 1. ($3 \times 2 + 1 = 7$), ($7 \times 2 + 1 = 15$). So, $63 \times 2 + 1 = 127$.
2. **Q:** Find the missing term: 2, 3, 5, 7, 11, ?, 17
   Solution: 13
   Explanation: Prime number series. Next prime after 11 is 13.
3. **Q:** Find the wrong term in the series: 8, 13, 21, 32, 47, 63, 83
   Solution: 47
   Explanation: Step method (differences): 5, 8, 11, 15, 16, 20. This is irregular. Let's look at double difference: 3, 3, 4, 1, 4. No. Let's rethink TCS logic. Differences should be 5, 8, 11, 14, 17, 20 (Double difference of 3). So 32 + 14 = 46 (not 47). 46 + 17 = 63. The wrong number is 47.
4. **Q:** Find the next term: BDF, CFI, DHL, ?
   Solution: EJO
   Explanation: First letters: B(+1) C(+1) D(+1) E. Second: D(+2) F(+2) H(+2) J. Third: F(+3) I(+3) L(+3) O.

5. **Q:** Find the missing term: 10, 5, 5, 7.5, 15, ?
   Solution: 37.5
   Explanation: Decimal Multiplication Trap. $10 \times 0.5 = 5$. $5 \times 1 = 5$. $5 \times 1.5 = 7.5$. $7.5 \times 2 = 15$. $15 \times 2.5 = 37.5$.

6. **Q:** Find the missing term: 0, 6, 24, 60, 120, ?
   Solution: 210
   Explanation: $n^3 - n$ trap! $1^3 - 1 = 0$, $2^3 - 2 = 6$, $3^3 - 3 = 24$, $4^3 - 4 = 60$, $5^3 - 5 = 120$. Next is $6^3 - 6 = 216 - 6 = 210$.

7. **Q:** Find the missing term: 8, 24, 12, 36, 18, 54, ?
   Solution: 27
   Explanation: Alternating operations. $\times 3$, then $\div 2$. $8 \times 3 = 24$, $24 \div 2 = 12$, $12 \times 3 = 36$, $36 \div 2 = 18$, $18 \times 3 = 54$, $54 \div 2 = 27$.

8. **Q:** Q1F, S2E, U6D, W21C, ?
   Solution: Y88B
   Explanation: First letter: Q(+2) S(+2) U(+2) W(+2) Y. Last letter: F(-1) E(-1) D(-1) C(-1) B. Middle number: $1 \times 1 + 1 = 2$, $2 \times 2 + 2 = 6$, $6 \times 3 + 3 = 21$, $21 \times 4 + 4 = 88$.

9. **Q:** Find the missing term: 121, 144, 289, 324, 529, 576, ?
   Solution: 841
   Explanation: Squares of alternating sequence of primes and composites. $11^2, 12^2, 17^2, 18^2, 23^2, 24^2$. The base numbers are (prime, prime+1). 11, 17, 23 are consecutive primes. Next prime is 29. So $29^2 = 841$.

10. **Q:** Find the wrong term: 1, 2, 6, 15, 31, 56, 91
    Solution: 91
    Explanation: Differences are 1, 4, 9, 16, 25... These are perfect squares ($1^2, 2^2, 3^2, 4^2, 5^2$). The next difference should be $6^2 = 36$. So $56 + 36 = 92$, not 91.

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3 Syllogism

Tricks, Steps & Shortcuts

- **Minimum Overlap Rule:** Always draw the Venn Diagram with the *least* possible intersection. Do not assume relations that are not explicitly stated.
- **Possibility Rule:** If no negative relationship (No / Some Not) is given between two entities, ANY possibility between them is True.

TCS Traps to Avoid

- **The "Only A Few" Trap:** TCS asks this 90% of the time. "Only a few A are B" means TWO things simultaneously: (1) Some A are B, AND (2) Some A are NOT B. It is a definite restriction.
- **The "Only A is B" Trap:** This translates to "All B is A", but with a lock! B cannot intersect with ANY other circle. It belongs strictly to A.
- **Either-Or Trap:** Check for three conditions: 1) Both conclusions must be individually false. 2) Elements must be the same (e.g., Cats/Dogs). 3) One positive (Some/All) and one negative (No/Some Not) conclusion.

Top 10 High Priority Questions

1. **Q:** Statements: All A are B. Some B are C.
   Conclusions: I. Some A are C. II. No A is C.
   Solution: Either I or II follows.
   Explanation: A is inside B. C overlaps B. We don't know if C overlaps A. Both conclusions are false individually. Elements are same. One is Some (+), one is No (-). Hence, Either-Or.

2. **Q:** Statements: Only a few Trees are Green. All Green are Plants.
   Conclusions: I. Some Trees are Plants. II. All Trees being Green is a possibility.
   Solution: Only I follows.
   Explanation: "Only a few Trees are Green" means Some Trees are Green and Some Trees are NOT Green. Therefore, All Trees can NEVER be Green (Conclusion II is false). Since Green is inside Plants, the Trees touching Green also touch Plants. So, I is True.

3. **Q:** Statements: No Car is a Bus. Some Buses are Trucks.
   Conclusions: I. Some Trucks are not Cars. II. All Trucks can be Cars.
   Solution: Only I follows.
   Explanation: The part of the Trucks that are Buses can NEVER be Cars. Therefore, Some Trucks are not Cars is definitely True. Because of this restricted part, All Trucks can never go inside Cars. So II is False.

4. **Q:** Statements: Only Dogs are Cats. Some Dogs are Rats.
   Conclusions: I. Some Cats are Rats. II. No Cat is a Rat.
   Solution: Only II follows.
   Explanation: Trap! "Only Dogs are Cats" means "All Cats are Dogs" AND Cats cannot interact with anything else. So Cats cannot touch Rats. Conclusion II is definitively true.

5. **Q:** Statements: Some A are B. Some B are C. Some C are D.
   Conclusions: I. Some A are D. II. All A being D is a possibility.
   Solution: Only II follows.
   Explanation: No direct relation between A and D is stated. Therefore, definite conclusions (I) are false, but possibility conclusions (II) are true.

6. **Q:** Statements: All Pens are Pencils. No Pencil is an Eraser.
   Conclusions: I. No Pen is an Eraser. II. Some Pencils are Pens.
   Solution: Both I and II follow.
   Explanation: Pen is entirely inside Pencil. If Pencil cannot touch Eraser, Pen definitely cannot. (I is True). "All Pens are Pencils" inherently means "Some Pencils are Pens" is true.

7. **Q:** Statements: Only a few M are N. Only a few N are O.
   Conclusions: I. Some M are O. II. All N can be M.
   Solution: Only II follows.
   Explanation: M and O have no direct link, so I is false. "Only a few M are N" means M restricts itself from going fully into N. BUT, N can fully go inside M! Thus, All N being M is a possibility.

8. **Q:** Statements: At least some Books are Novels. All Novels are Poems.
   Conclusions: I. All Books can be Poems. II. Some Poems are Books.
   Solution: Both I and II follow.
   Explanation: "At least some" just means "Some". Books and Novels overlap. Novels are fully inside Poems. Therefore, the overlap is shared with Poems. II is true. Since there are no negative statements, any possibility is true (I is true).

9. **Q:** Statements: 100% A are B. 0% B are C.
   Conclusions: I. Some A are not C. II. Some B are A.

Solution: Both I and II follow.
Explanation: 100% = All. 0% = No. All A are B, No B is C. Therefore No A is C. ”No A is C” implies ”Some A are not C” is also true. ”All A are B” implies ”Some B are A”.

10. **Q:** Statements: Some Reds are Blues. All Blues are Blacks. No Black is White.
Conclusions: I. Some Reds are not Whites. II. No Blue is White.
Solution: Both I and II follow.
Explanation: The part of Red that is Blue is also Black. Since No Black is White, that specific part of Red can never be White. So I is true. Blue is entirely inside Black, and Black cannot touch White. So Blue cannot touch White. II is true.

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4 Directions

Tricks, Steps & Shortcuts

- **Algebraic Degree Tracking:** Do not draw lines for purely rotational questions. Face North = 0 degrees. Right = Clockwise (+), Left = Anti-clockwise (-). Just add/subtract the angles.
- **Pythagoras Triplets:** For shortest distance, memorizing triplets saves 20 seconds. (3,4,5), (5,12,13), (6,8,10), (8,15,17), (7,24,25).
- **NESW Write-Down Rule:** Write N - E - S - W left to right. A right turn moves you right on the list (E to S). A left turn moves you left (S to E).

TCS Traps to Avoid

- **The Shadow Trap:** ”Morning” means Sun is in the East, shadows fall West. ”Evening” means Sun is West, shadows fall East. If a person’s shadow is to his ”Right” in the morning, since shadow is West, his Right is West. Therefore, he is facing South.
- **The Reversal Trap:** ”Walks backward” or ”Starts from end point”. Read the last sentence first to see if they want starting direction or ending direction.

Top 10 High Priority Questions

1. **Q:** A man faces North. He turns 135 degrees clockwise and then 45 degrees anti-clockwise. Which direction is he facing now?
Solution: East
Explanation: Clockwise = +135. Anti-clockwise = -45. Net movement = +135 - 45 = +90 degrees. 90 degrees clockwise from North is East.

2. **Q:** A person walks 5 km North, turns right and walks 12 km. What is the shortest distance from the starting point?
Solution: 13 km
Explanation: Forms a right-angled triangle. Hypotenuse = $\sqrt{5^2+12^2}=\sqrt{25+144}=\sqrt{169}=13$. (Or just use the 5,12,13 triplet).

3. **Q:** One morning after sunrise, Vimal started walking. His shadow fell exactly to his right. Which direction is he facing?
Solution: South
Explanation: Morning = Sun is in the East. Shadow is in the West. If West is to his Right, he must be facing South.

4. **Q:** A runs 10m East, turns Right and runs 10m, turns Right again and runs 10m. How far is he from his starting point and in which direction?
   Solution: 10m, South
   Explanation: East $\rightarrow$ Right = South $\rightarrow$ Right = West. He went 10m East, 10m South, 10m West. The East and West 10m cancel out. He is 10m South of the start.

5. **Q:** Point A is 5m West of B. Point C is 5m South of B. Point D is 5m East of C. In which direction is D with respect to A?
   Solution: South-East
   Explanation: A is West of B. C is South of B. So C is South-East of A. D is East of C. Therefore, D is deeply into the South-East quadrant relative to A.

6. **Q:** A girl leaves her house, walks 30m North, turns North-West and walks 30m, then turns South and walks 30m, then turns South-East and walks 30m. How far is she from her house?
   Solution: 0m (She is at her house)
   Explanation: This traces a perfect rhombus/parallelogram shape returning to the origin point.

7. **Q:** At 6:00 PM, the hour hand of a clock points North. In which direction will the minute hand point at 9:15 PM?
   Solution: West
   Explanation: At 6:00, the hour hand is at 6. If 6 is North, then the clock is upside down (12 is South, 3 is West, 9 is East). At 9:15, the minute hand is at 3. The position 3 corresponds to West.

8. **Q:** One evening before sunset, Rekha and Hema were talking to each other face to face. If Hema's shadow was exactly to the right of Hema, which direction was Rekha facing?
   Solution: South
   Explanation: Evening = Sun West, Shadow East. Hema's shadow is to her right. So East is Hema's right. Therefore, Hema is facing North. Since they are face to face, Rekha is facing South.

9. **Q:** I am facing South. I turn right and walk 20m. Then I turn right again and walk 10m. Then I turn left and walk 10m. Then I turn right and walk 20m. In which direction am I from the starting point?
   Solution: North-West
   Explanation: Trace it: Face South. Right = West (20m). Right = North (10m). Left = West (10m). Right = North (20m). Total movement: West = 20+10 = 30m. North = 10+20 = 30m. Result: 30m North, 30m West $\rightarrow$ North-West.

10. **Q:** A man walks 1 km East, turns South and walks 5 km. Again he turns East and walks 2 km. Finally, he turns North and walks 9 km. How far is he from his starting point?
    Solution: 5 km
    Explanation: Total East = 1 + 2 = 3 km. Net North/South = Went 5 km South, then 9 km North = 4 km North. Now he is 3 km East and 4 km North of start. Distance = $\sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$ km.